Post-Training for Embodied Foundation Models: SFT and RL

SFT and RL for VLA / WAM

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Course repository ·WAM-RL paper ·RLinf documentation ·简体中文 ·PDF

Item	Details
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1. What is post-training?

Supervised Fine-Tuning and Reinforcement Learning for Vision–Language–Action Models and World Action Models
Objectives, trainable interfaces, implementation boundaries, and deployment ·October 2026

> **Scope.** This course starts from **pretrained models** and distinguishes supervised fine-tuning (SFT) from reinforcement learning (RL) post-training. It examines VLA and WAM architectures separately. The VLA RL material retains the six methods and RLinf infrastructure discussion from the original tutorial; the WAM part follows Qian et al., *WAM-RL* (2026). Low-level controller training from scratch and executable labs are outside the scope.
>
> **Evidence boundary.** Benchmark outcomes apply to the specified model and evaluation settings. Neither a WAM-RL result nor support for different components in one repository implies universal WAM superiority or plug-and-play compatibility. Extension proposals are distinguished from demonstrated capabilities; no independent reproduction or physical-robot validation is claimed.
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Pretraining builds broad visual, linguistic, and motor priors. **Post-training** adapts an existing model with weights θ_0 to specific tasks, robot embodiments, data distributions, or feedback. SFT versus RL describes the **learning signal**; VLA versus WAM describes the **architecture**. They are orthogonal axes.

Architecture	SFT: supervised target data	RL: environment interaction and reward
VLA	Fit expert actions by updating an action expert, backbone, or adapter	Optimize the generator, or freeze it and learn noise-, feature-, or action-level actors
WAM	Learn future-video/world-state prediction and the video-to-action mapping	Improve the actor's execution fidelity; WAM-RL also adapts the world model through online Video SFT

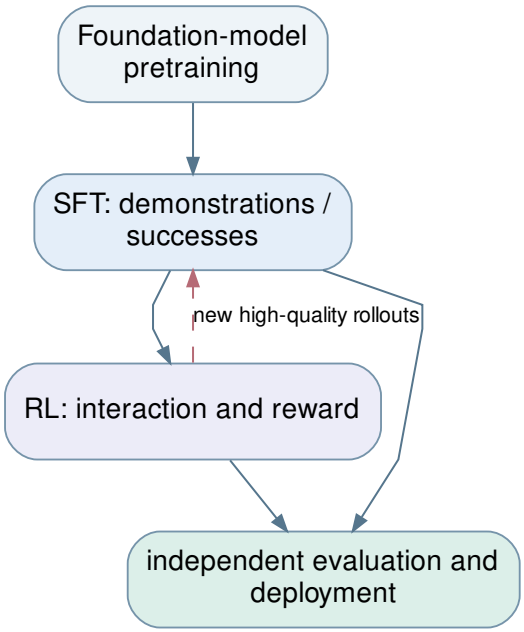


Figure 1. SFT and RL can alternate or be interleaved. SFT followed by RL is a common recipe, not a universal sequencing requirement.

Let o_t denote observation, g_t the goal or language instruction, and $A_t = (a_t, \dots, a_{t+H-1})$ a candidate action chunk of length H. The number of low-level steps actually executed before replanning is K, which need **not equal** H. For either architecture, we must identify the supervised target, parameter update scope, reward origin, replay semantics, and skill-preservation constraints.

2. SFT and RL objectives

2.1 SFT: predict the demonstrated action or future

For an explicit conditional action distribution, the VLA behavior-cloning objective is

$$\mathcal{L}_{\text{SFT}}^{\text{VLA}}(\theta) = -\mathbb{E}_{(o,g,A)\sim\mathcal{D}_{\text{sup}}} \log \pi_\theta(A \mid o, g).$$

Diffusion and flow action experts usually use a **denoising or velocity-matching objective**, rather than the above likelihood in its naive form. A representative conditional flow-matching loss for $x_\tau = (1-\tau)x_0 + \tau x_1$ is

$$\mathcal{L}_{\text{FM}} = \mathbb{E}_{x_0,x_1,\tau,c}\left[\|v_\theta(x_\tau, \tau, c) - (x_1 - x_0)\|_2^2\right],$$

where $c = (o, g)$ and x_1 is the target action chunk or future-video sample. Other probability paths and flow parameterizations exist. SFT may specialize a pretrained policy to a domain, a robot, or a skill; SFT on **new successful rollouts during RL** is also post-training.

2.2 RL: learn from the quality of outcomes

The standard discounted-return objective is

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{T}\gamma^t r_t\right].$$

SFT answers, "What should the model imitate?" RL answers, "Which behavior produces more return in the environment?" RL can improve beyond the demonstration support but requires useful exploration, credible reward, suitable value estimates, and interaction. Imitation constraints, reference-policy regularization, and periodic SFT may mitigate forgetting during RL.

Do not conflate Online with On-policy. Online means new environment data are collected during learning. Off-policy means the learner may also reuse transitions from earlier behavior policies. Likewise, Online Video SFT inside an RL training loop does not mean the world model receives a policy-gradient update.

3. VLA SFT post-training

3.1 Which parameters are updated?

A typical VLA pipeline is **image + language + robot state → visual-language representation → action chunk**. An action-token head commonly uses cross-entropy, whereas a continuous diffusion/flow expert typically uses denoising/flow matching. These are not interchangeable log-probability interfaces.

Parameterization	What changes?	Main cost or risk
Full fine-tuning	All selected pretrained parameters	High memory and compute; representation and skill drift
LoRA / adapters	Low-rank or additional modules	Limited adaptation capacity or unsuitable insertion points
Action Expert only	Motor-generation network; backbone remains fixed	Cannot fix all vision/language domain gaps
Frozen backbone + new head	An embodiment-specific action interface	Requires correct action semantics and time scales

3.2 Data, time alignment, and evaluation

A useful supervised sample includes (o_t, g_t, A_t) along with embodiment information, action units (joint versus end-effector versus finger commands), normalization, acquisition rates, camera calibration, and timestamps. Multi-task SFT requires a sensible mixture across robots, instructions, successes, and recovery segments. **Only imitating idealized successful demonstrations is often insufficient for recovery**.

The practical logic is: **multimodal data → spatial/temporal alignment → chunked supervision → selective fine-tuning → independent closed-loop evaluation**. SFT can be connected to RL only if observation, executed action, termination, reward, and model-version semantics agree. A lower supervised loss is not a substitute for real task success or contact safety.

4. VLA RL post-training: six routes retained from the original tutorial

A major obstacle is that a diffusion/flow policy may not expose a tractable **marginal action log-probability or policy KL**. Ordinary Gaussian PPO/SAC cannot be directly bolted onto an arbitrary generative head. The first organizing question is **what actually receives the RL update?**

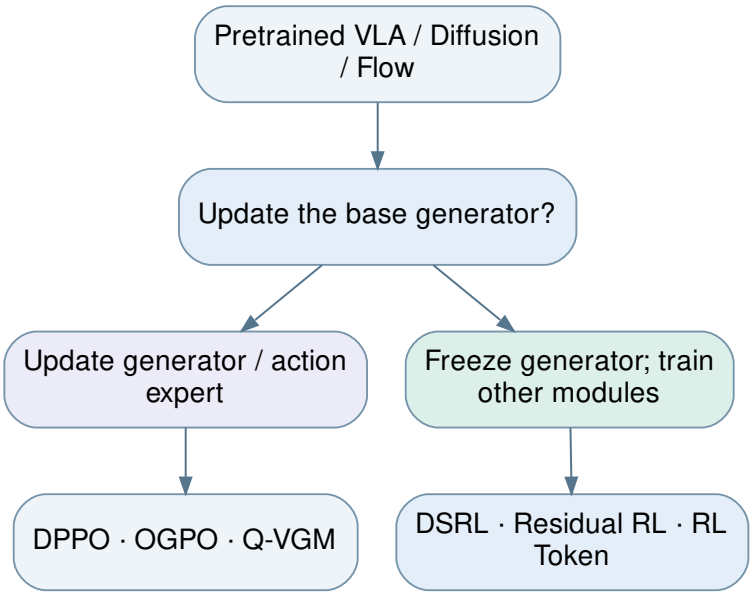


Figure 2. Alternative trainable interfaces, not six sequential optimization stages. LoRA and hybrid approaches can cross the two branches.

4.1 Directly update the generator or action expert

DPPO — diffusion policy gradients. DPPO incorporates the denoising chain into a policy-optimization formulation instead of pretending the final diffusion action is a simple Gaussian policy. PPO's clipped surrogate provides intuition:

$$L_{\text{clip}} = \mathbb{E}\left[\min\left(\rho\hat{A}, \text{clip}(\rho, 1-\epsilon, 1+\epsilon)\hat{A}\right)\right].$$

This is **not DPPO's complete loss**. On-policy rollouts, repeated denoising, gradient cost, and potential degradation of pretrained skills matter.

OGPO — off-policy replay for generative policy improvement. OGPO learns an off-policy critic from older and new trajectories and passes value information to the generator: **replay → critic/advantage → generator update**. PPO-like loss components do not make the algorithm on-policy PPO. Critic bias, behavior-policy shift, and long generation-chain optimization are important.

Q-VGM — Q-guided flow matching. Q-VGM constructs a local value-guided velocity-matching objective for the Flow Action Expert without backpropagating through the full generation trajectory. Its value learning combines offline IQL and online TD estimation. **The pretrained generator is updated** in this route.

4.2 Freeze the generator and optimize auxiliary modules

DSRL — noise/latent actor. DSRL learns a smaller proposal policy while keeping the generator fixed:

$$z \sim q_\phi(z \mid o, g), \qquad A = G_{\theta_0}(o, g, z), \quad \theta_0 \text{ fixed.}$$

This preserves the base generator and reduces trainable parameters, but reachable behavior remains constrained by the

frozen generator's action support. A noise interface validated for a specific diffusion model cannot automatically be transferred to all VLA/flow models.

Residual RL — restricted corrections in action space. Let the frozen policy produce A_0 and a small actor produce ΔA_ϕ:

$$A_{\text{cmd}} = \text{Safe}(A_0 + \lambda \Delta A_\phi).$$

Residual RL is an **architecture**, not one specified on- or off-policy algorithm. Safe action scaling, coordinate frames, chunk boundaries, and contact stability matter, especially for dexterous manipulation.

RL Token — lightweight feature-based actor–critic. RL Token constructs a compact VLA feature interface and then supports online off-policy actor–critic updates with the main representation network frozen. Creating this feature interface may itself require demonstration-based adaptation. It should not be confused with an action residual.

4.3 VLA RL method-selection table

Method	RL update target	Common data setup	Main constraint
DPPO	Diffusion generator	Mostly on-policy	High rollout/denoising cost
OGPO	Generative policy	Online off-policy	Replay drift and critic reliability
Q-VGM	Flow velocity field	Offline-to-online	Accurate local Q gradients
DSRL	Noise/latent actor	Can be off-policy	Frozen generator, limited action support
Residual RL	Action-correction actor	Algorithm-dependent	Chunk alignment and contact safety
RL Token	Feature actor–critic	Online off-policy	Feature compatibility and preliminary adaptation

First choose a **trainable interface**, then an **optimization objective and data regime**. Full/partial updates, velocity matching versus policy gradients, and on-policy versus off-policy are distinct classification axes.

5. WAM architecture and SFT post-training

5.1 World prediction and executable action are distinct roles

In the **World–Action** architecture studied by WAM-RL, a world model forecasts future visual observations or their latent representations, and an actor maps the predicted information to motor commands:

$$\underbrace{\hat{X}_{t+1:t+H},\ z_t = f_\theta(X_{\le t}, g)}_{\text{World Model}} \quad \longrightarrow \quad \underbrace{A_t = \pi_\phi(z_t, o_t, g)}_{\text{Action Model}}.$$

X denotes a video/observation sequence. Not every model labeled "WAM" has identical modules; this is the **working decomposition for WAM-RL**. It explicitly couples "what the future might look like" and "which actions realize it."

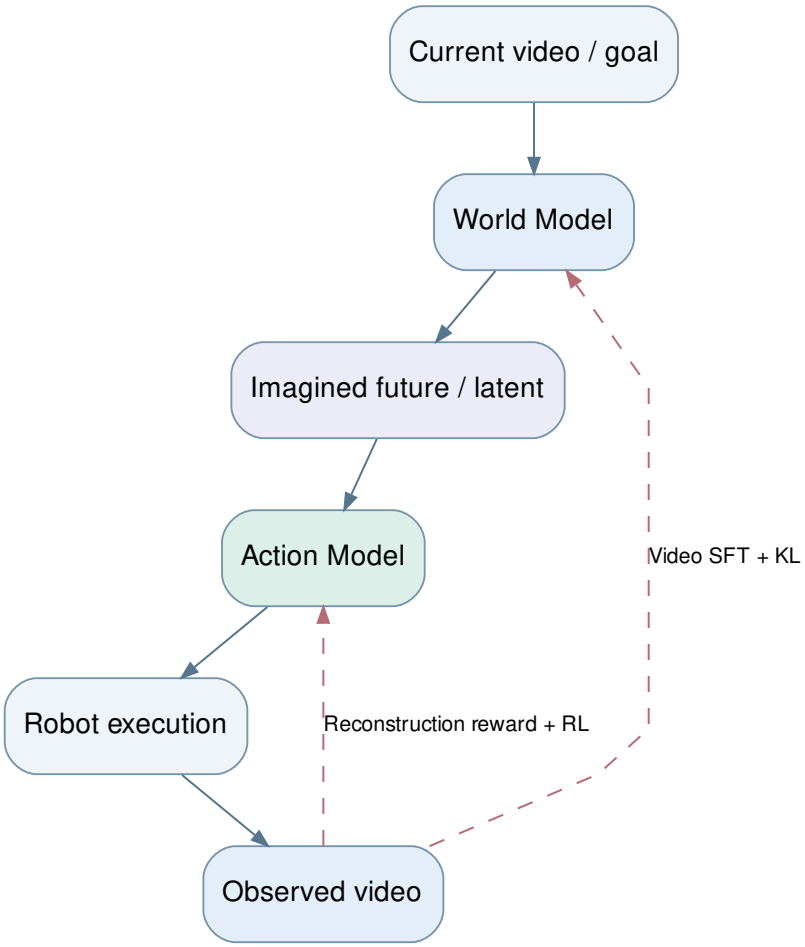


Figure 3. WAM-RL combines online Video SFT for the world model and reconstruction-reward RL for the actor.

5.2 Two possible supervised interfaces

General WAM SFT can include **video prediction/world dynamics supervision** and **action mapping supervision**, or a joint video-and-action generative objective. Consequently, "WAM SFT only trains videos" is not a universal claim. The specific **online post-training** recipe in WAM-RL adapts the world model using successful rollout videos while optimizing the actor through RL.

The paper's video objective is

$$\mathcal{L}_{\text{video}} = \mathbb{E}_{X_{1:T}\sim\mathcal{D}_{\text{success}}}\big[\ell(f_\theta(X_{<t}), X_t)\big],$$

where ℓ can be a reconstruction or flow-matching loss. A trajectory labeled ultimately successful may nevertheless contain a failed grasp followed by a corrective retry: such segments expose the model to **recovery dynamics**. Successful-only filtering still leaves many failure modes underrepresented.

5.3 Latent KL regularization to protect actor compatibility

The actor depends on features produced by the world model. Updating the world model can shift those features and destabilize an actor trained on the old representation. WAM-RL constructs **diagonal Gaussian proxies** for deterministic latent features:

$$q_\theta(z_t) = \mathcal{N}(z_t, \Sigma_\theta), \qquad q_0(z_t) = \mathcal{N}(z_t^0, \Sigma_0),$$

with $z_t = f_\theta(X_{<t})$, $z_t^0 = f_{\theta_0}(X_{<t})$, a frozen reference world model θ_0, an EMA estimate of current feature-scale covariance, and a fixed reference covariance. **The Gaussian assumption is a tractable surrogate**, not a claim that these deterministic features naturally follow known distributions.

$$\mathcal{L}_{\text{WM}} = \mathcal{L}_{\text{video}} + \lambda_{\text{KL}} \mathbb{E}_t[D_{\text{KL}}(q_\theta(z_t) \| q_0(z_t))].$$

The analytical Gaussian KL penalizes abrupt feature drift and helps the world model and actor remain compatible. The tradeoff is that stronger regularization can limit the world model's ability to depart from the pretrained distribution.

6. WAM RL post-training: the WAM-RL case study

6.1 Reward: realize the imagined future

The world model predicts $\hat{X}_{t+1:t+H}$. Actions are executed to obtain $X_{t+1:t+H}$. WAM-RL builds a dense reconstruction reward:

$$r_t^{\text{recon}} = \text{sim}(\hat{X}_{t+1:t+H}, X_{t+1:t+H}).$$

The study compares pixel MSE, optical-flow MSE, DINOv2 feature error, and V-JEPA2 feature error. A negative pixel MSE is one natural similarity score. **Reconstruction consistency is not itself task success**: executing an incorrect world-model prediction faithfully does not guarantee satisfying the user's objective. The quality and task relevance of the imagined plan therefore constrain this reward.

6.2 RL for a flow-based actor via Flow-SDE

A deterministic flow ODE is not automatically a stochastic policy with a convenient action likelihood. WAM-RL uses a Flow-SDE construction to introduce noise and tractable sampling transitions:

$$dx_\tau = v_\phi(x_\tau, \tau)\, d\tau + \sigma\, dW_\tau,$$

$$p_\phi(x_{i-1} \mid x_i) = \mathcal{N}(x_{i-1}; \mu_\phi(x_i, i), \sigma^2 I).$$

The log-probabilities of a **sampled path** can then be accumulated for policy-gradient optimization:

$$\nabla_\phi J \approx \mathbb{E}[\nabla_\phi \log p_\phi(\text{sampling path} \mid s_t)\, \hat{A}_t].$$

This deliberate distinction matters: a joint sampling-path density should not automatically be equated with the exact marginal log-probability of the final action. Implementations must optimize the quantity justified by their stochastic generation formulation; generic PPO on any deterministic flow head is not sufficient.

6.3 Coordinated updates, not a single RL loss for everything

The conceptual training loop consists of four activities:

1. **Predict**: the world model generates an imagined future and the actor maps its representation to actions.
2. **Interact**: execute actions and record synchronized video, action, and task outcomes.
3. **Actor RL**: compare imagined and executed video, compute dense reward/advantages, and update the action model.
4. **Online World Model Video SFT**: use successful trajectories to refine predictions while keeping latent statistics close to a reference through KL.

Thus, WAM-RL is **joint post-training/co-adaptation**, not the claim that policy gradients update every component. Its world-model update is supervised; its actor update is reinforcement-based.

6.4 Reported results and their limits

The following are the paper's **reported success rates in specified benchmark settings**:

Method	LIBERO-Object	RLBench Water Plants
Pretrained WA base	68%	19%
Actor-only RL (π_{RL})	78%	18%
WAM-RL (actor RL + online world-model SFT)	**82%**	**22%**

These results support the claim that actor-only learning may be limited when long-horizon errors are dominated by the predictive world model. They do **not** establish universal WAM superiority to VLA, or universal post-training superiority to all task-specialized alternatives. The authors implement WAM-RL on **Genie Envisioner-ACT** and report **eight NVIDIA A800 GPUs for eight hours** for their training setting; that configuration is not a general minimum requirement for WAMs.

In a **separate reward ablation setting** on RLBench Water Plants, Pixel MSE, Optical Flow MSE, DINO MSE, and V-JEPA2 yield **21%, 19%, 16%, and 17%**, compared with 19% for base and 18% for actor-only in that table. The ablation's Pixel MSE **21% must not be silently substituted for the main table's WAM-RL 22%**. Greater separation between successful and failed trajectory rewards (optical flow) did not translate into the highest task success; alignment with the world-model objective also matters.

The paper explicitly notes two limitations: KL-constrained video adaptation may restrict capability growth, and reconstruction rewards based on hand-designed metrics or pretrained features may insufficiently discriminate task success from failure. Scalable video updating, safe real-robot learning, reliable contact rewards, and long-term cross-embodiment adaptation remain research questions.

7. Systematic VLA versus WAM post-training comparison

Dimension	VLA post-training	WAM post-training (WAM-RL as example)
Architecture	Visual-language features → action head/chunk	World prediction → latent → action model
SFT target	Expert action tokens or continuous generative action loss	Future video/dynamics and optionally supervised action mapping
RL update	Generator, adapter, noise actor, feature actor, or residual	Actor execution fidelity; world model adapted with Video SFT
RL reward	Task success, progress, contact, learned value, etc.	Imagined-versus-realized visual reconstruction consistency
Stability concern	Policy likelihood, critic bias, forgetting	These issues plus world/actor latent mismatch

Dimension	VLA post-training	WAM post-training (WAM-RL as example)
Inference compute	Depends on VLM/expert and sampling steps	Typically adds future generation, depending on architecture/horizon
Training compute	Model gradients plus rollouts/replay	Additionally maintains/updates a video world model
Hypothesized advantage	Direct action-generation interface	Explicit future prediction may support recovery and long-horizon decisions
Principal risk	Contact precision, OOD behavior, chunk drift	Prediction errors, reward misalignment, latent drift, video latency

Compute claims require matched conditions. Compare parameter counts, frame resolution, predicted horizon, denoising steps, batch size, hardware, GPU-hours, robot-hours, and actual success gains. A single paper's 8×A800 result does not provide a fair VLA-versus-WAM compute ranking. A high success rate on one benchmark does not imply that a WAM always needs less SFT/RL than a VLA.

8. Shared online RL and actor–critic semantics

An off-policy experience store can contain $\mathcal{D} = \{(o_t, A_t, r_t, o_{t+K}, d_{t+K}, K)\}$. A **pedagogical** chunk-aware Bellman target is

$$y_t^{(K)} = \sum_{j=0}^{K-1} \gamma^j r_{t+j} + \gamma^K (1 - d_{t+K}) \mathbb{E}_{A' \sim \pi}[Q_{\bar{\psi}}(o_{t+K}, A')].$$

The generated action chunk length H and physically executed length K are not the same variable. Replay, discounting, timestamps, chunk overrides, and critic targets should refer to executed steps. This is an illustrative Bellman expression, **not the full objective of SAC, OGPO, Q-VGM, or WAM-RL**.

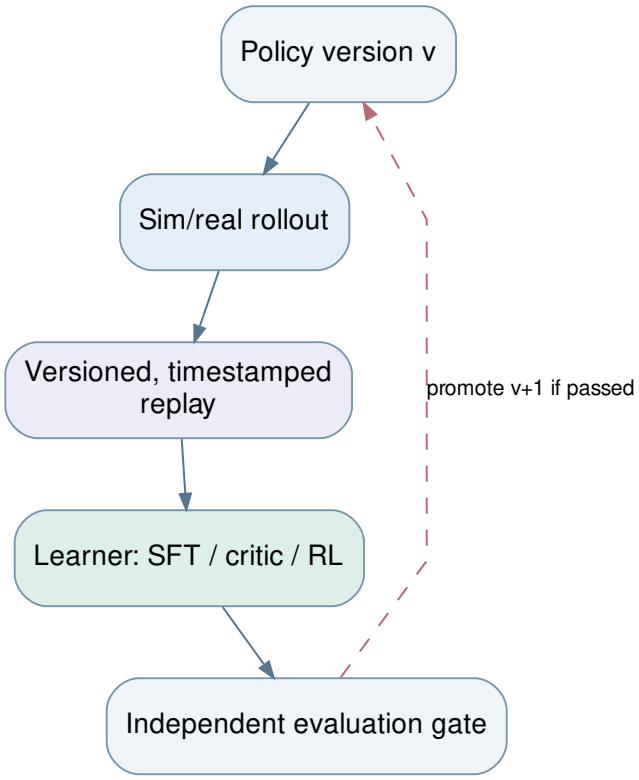


Figure 4. Model synchronization, data semantics, and independent deployment gates are part of post-training, not guaranteed by a learning loss.

9. RLinf infrastructure: retain VLA capabilities, identify WAM integration requirements

RLinf provides components for model training, rollout scheduling, actor–learner workflows, SFT/RL, evaluation, and several physical-robot pathways. It is a useful infrastructure example, **not proof of universal compatibility** across arbitrary models, algorithms, and robot hardware.

9.1 Documented VLA-side recipes and limitations

Public recipe / capability	What it supports	What it does not establish
$\pi_0/\pi_{0.5}$ PPO/GRPO	Specific on-policy VLA RL settings	General off-policy adaptation of any flow VLA
DSRL with frozen π_0	Noise-actor updates while base policy is frozen	Drop-in noise interface for any new model
SAC-Flow	SAC on a dedicated flow actor	Verified frozen-$\pi_{0.5}$ SAC training
RLT / RL Token	Feature preparation and lightweight actor–critic	Residual RL without representation adaptation
Real-robot guidance	Examples of real-data and multi-robot workflows	Automatic adaptation to arbitrary hands/arms

Public recipe / capability	What it supports	What it does not establish
RTC	Overlap of policy inference and robot execution	Hard real-time guarantees or safe motor control
Adding a Robot	Extension interfaces for hardware and tasks	Universal ready-made robot SDK integration

It is incorrect to say that RLinf supports **only RL**, or that it categorically lacks **off-policy training or real robots**. It is equally incorrect to infer that **WAM-RL is already integrated end-to-end** merely because RLinf has VLA, SFT, and RL capabilities. A new WAM interface would require verification of future-video supervision, imagined and realized video alignment, latent extraction, frozen reference/KL statistics, actor gradients, and multi-model resource scheduling.

9.2 Incremental engineering requirements for WAM-RL

Subsystem	Frequent VLA requirement	Additional WAM-RL requirement
Data	Observation, executed action, reward, termination	Imagined video, realized video, successful-rollout filtering
Learners	Expert/actor and critic optimization	Coupled video SFT and actor RL updates
Representation	Compatible action and vision features	Stable world-model latent contract; reference/KL
Reward	Task completion, progress, contact	Time-aligned reconstruction similarity
Inference	Action chunks and perhaps RTC	Future generation, latency/cache/memory budget
Evaluation	Skill success, recovery, regression	Prediction error, execution match, actual task outcome

This table is an **integration checklist inferred from WAM-RL**, not a source-code audit claiming each component is absent from RLinf.

9.3 Retain the original RLinf training-side limitations

Gap	Why it does not disappear automatically	Checks for a new model
Model × algorithm compatibility	$\pi_{0.5}$ + PPO/GRPO, frozen π_0 + DSRL, and dedicated Flow + SAC are distinct recipes	Path likelihood, gradients, parameter freezing, sampling noise
Generic online off-policy flow/VLA	Experience reuse requires a critic and a compatible generative-policy update	Updated tensors, policy lineage, replay mixture
Contact-aware critic reliability	Occlusion, sparse feedback, and OOD replay can amplify Q errors	Q calibration, failures, success labels, risk constraints
Action chunk semantics	Generated H and executed K may differ	K-step return, γ^K, termination, timestamps, overrides
Compute and throughput	VLA inference, environment throughput, distributed learners, and robot resets all matter	GPU-/robot-hours, worker backpressure, checkpoints, reproducibility
Multi-task skill preservation	Task-specific online RL may damage old behaviors	Held-out regression, SFT/replay mixing, rollback

These gaps span **limits of validated model/algorithm combinations** and **open reliability problems**. They must not be collapsed into the claim that RLinf simply lacks a code feature. A WAM additionally requires dual-learner coordination, latent compatibility, and trustworthy reconstruction rewards.

10. Real-robot deployment and unresolved questions

A training reward is not a safety controller. Simulated rollouts must not be silently identified with physical interactions, and large-model inference is distinct from high-frequency servo control. For a platform with **manipulator + dexterous hand + head/wrist cameras**, coordinate calibration, sensor timestamps, chunk arbitration, finger safety, and reset/recovery are first-class requirements.

Risk domain	Independent checks and metrics
Timing	Mean/P95/P99 latency, network jitter, chunk discontinuities, servo frequency
Runtime safety	Joint/current limits, collision checks, watchdog, emergency stop, rollback
Reward validity	Independent success evaluator; false positives and correlation with reconstruction reward
Continual learning	Replay/model lineage, old-skill regression, GPU/robot-hours, reset overhead
New robot integration	Vendor SDK, CAN/RS485/ROS, action units, reset and failure recovery
Multimodal perception	Calibration, camera clocks, occlusion, reward uncertainty, OOD detection
Deployment versioning	Independent candidate-policy gating, safe rollout and rollback
WAM-specific	Prediction horizon versus execution delay, latent drift, misleading imagined futures

Avoid five incorrect equivalences: **Online = On-policy**; **Frozen generator = no training**; **Velocity matching = full fine-tuning**; **PPO/GRPO examples = online off-policy recipes**; **Simulated success = real-robot safety**. None is valid.

Research opportunities include **data-efficient real-robot online off-policy VLA updates**, **expanding control capability beyond frozen generator support**, **contact-aware critics**, **stable WAM world/actor co-adaptation**, **long-horizon video prediction under real-time constraints**, and **multi-robot continual learning**. None follows automatically from a documented training recipe.

11. Decision guide

Starting point	Primary post-training route	What to verify
VLA lacks demonstration-based task competence	**VLA SFT**	Action semantics, head objective, prior-skill retention
VLA has a skill but low task success	**VLA RL**: DPPO / OGPO / Q-VGM / DSRL / Residual / RL Token	Critic quality, rollout cost, executable chunk
WAM predicts inaccurate futures	**World Model SFT**, possibly with online successful videos	Prediction fidelity and actor latent compatibility
WAM predicts adequately but acts unreliably	**Actor RL**, e.g., reconstruction reward	Time alignment, genuine task outcomes
WAM suffers from long-horizon prediction drift	**Coordinated World Model Video SFT + Actor RL**	KL strength, synchronized updates, closed-loop success

Takeaway. SFT learns from high-quality targets; RL improves policy performance using feedback. VLA post-training primarily exposes **action-generation and auxiliary actor interfaces**. WAM-RL explicitly coordinates **world prediction**

and executable action, while preserving the difference between a supervised world-model update and a reinforcement-based actor update.

12. Primary papers and official resources

Topic	Resources
Foundations	PPO ·SAC ·Diffusion Policy
Update the generator	DPPO ·OGPO ·Q-VGM
Frozen/auxiliary actor	DSRL ·RL Token
WAM and post-training	DreamZero ·WAM-RL
Training/deployment infrastructure	RLinf ·RLinf-USER ·RTC

Materials checked through 20 October 2026. Mathematical formulations are pedagogical unless explicitly attributed; research results retain the original benchmark context.